

Computational Linguistics – Project Report

Do linguistically motivated regex rules for formal Hindi broadcast speech generalize across registers and what do the failure modes reveal about register-specific variation?

mew not

Vidushi Agarwal · Ananya Pai
2025114001 · 2025114015

Computational Linguistics I
May 2026

Abstract

This report presents a computational linguistics study comparing a custom regex-based pipeline against a Stanza neural pipeline for extracting seven linguistically significant patterns from a mixed-script Hindi corpus. The primary dataset consists of transcripts from Mann Ki Baat, a formal radio broadcast (~9,800 tokens, three episodes: December 2025 – February 2026). A complementary informal corpus of ~4,200 tokens scraped from YouTube comments on the same videos was added to enable cross-register analysis. The seven patterns targeted are: reduplication, conjunctive verbs, compound verbs, honorifics, acronyms, multi-word expressions (MWEs), and hyphenated pairs. Both pipelines were evaluated against a manually annotated gold standard of approximately 725 sentences. Key findings include: (1) the neural pipeline outperforms regex overall (F1: 0.86 vs 0.59 on comments, 0.49 vs 0.40 on broadcast), yet (2) regex is competitive or superior for conjunctive verbs and acronyms in both registers; (3) MWEs and hyphenated pairs are strong register markers with formal-to-informal frequency ratios of 2.69× and 9.67× respectively; and (4) reduplication shifts pragmatic function from grammatical device in formal speech to rhetorical emphasis in informal social media Hindi.

1. Introduction

The Mann Ki Baat radio programme code-switches mid-sentence between Devanagari script, ISO 15919-transliterated Hindi, and English while simultaneously employing Sanskritized vocabulary and elaborate postpositional constructions. Standard NLP models fail silently on this register because no existing benchmark covers it.

This project addresses that gap by implementing and evaluating two parallel extraction pipelines, a linguistically motivated regex system and a Stanza-based neural system, on two corpora representing contrasting registers of the same topical domain. The central research question is: do linguistically motivated regex rules for formal Hindi broadcast speech generalize across registers, and what do the failure modes reveal about register-specific variation?

1.1 Motivation

Pattern identification in natural language processing underpins downstream applications ranging from information extraction to automatic summarization. For Hindi, whose rich agglutinative morphology and frequent code-switching create a distinctly challenging environment, the question of whether expensive neural models are necessary or whether cheaper rule-based systems suffice is practically significant. This study provides an empirical answer across two registers.

1.2 Research Objectives

- **Objective 1 — Cross-Register Pattern Distribution:** Construct a comparative frequency table of seven linguistic patterns across the formal and informal corpora, identify which patterns are register-stable versus register-sensitive, and determine whether informal social media Hindi preserves, suppresses, or transforms each pattern relative to broadcast speech.
- **Objective 2 — System Comparison (Regex vs. Stanza):** Evaluate rule-based regex extraction against a Stanza-based neural pipeline on both corpora independently, reporting per-pattern precision, recall, and F1. Determine whether hand-crafted rules remain competitive with a neural system on well-formed formal text, and how gracefully each approach degrades on noisy informal input.
- **Objective 3 — Lightweight Register Markers:** Determine whether any subset of the seven patterns functions as a reliable, computationally cheap signal for register classification without requiring a trained classifier.

2. Dataset

2.1 Formal Corpus: Mann Ki Baat Transcripts

The primary corpus consists of verbatim transcripts of three consecutive episodes of Mann Ki Baat (December 2025, January 2026, February 2026). This monthly radio programme is a formal broadcast speech genre, predominantly in Devanagari-script Hindi, with systematic English code-switching for technology terms, global-event proper nouns, and borrowed vocabulary.

Episode	Tokens	Sentences	% Devanagari
December 2025	3,321	208	69.0%
January 2026	3,268	214	70.3%
February 2026	3,174	211	70.9%
TOTAL	9,763	633	70.1%

The remaining ~30% of tokens in each episode include English code-switched words (tech terms, leader names), digit strings, and punctuation, a distribution rarely represented in standard Hindi NLP benchmarks.

2.2 Informal Corpus: YouTube Comments

To enable cross-register analysis, a complementary informal dataset of ~4,230 tokens (approximately 112 sentences) was scraped from YouTube comments on Mann Ki Baat videos. Constraining the comments to the same topic domain ensures that frequency differences between corpora reflect register rather than topic. A primitive normalization pipeline was applied to remove emojis and non-Hindi/English content before analysis.

The informal corpus exhibits higher lexical density of colloquial forms, non-standard orthography, abbreviated structures, and political discourse markers absent from the formal transcripts. It contains zero poorna viram (।) sentence-boundary markers, requiring purely whitespace-based sentence segmentation.

2.3 Preprocessing Pipeline

Both corpora underwent the following pipeline before pattern extraction:

- Unicode NFC normalization to canonicalize Devanagari ligatures
- UTF-8 validation (no invalid code points were found in either corpus)
- Punctuation normalization: smart-quote stripping, danda standardization
- ISO 15919 transliteration (Devanagari → Latin) for the regex pipeline, using a custom `unicode_converter.py` module
- Schwa deletion (word-final only): a known limitation is that medial schwas remain undeleted, producing e.g. `sarakār` instead of expected `sarkār`

3. Methodology

3.1 Linguistic Patterns Targeted

Seven patterns were selected based on their linguistic significance in Hindi and their expected relevance to register variation:

Pattern	Description
Reduplication	Hyphen-joined word repetitions (e.g. dhīrē-dhīrē, alag-alag)
Acronyms	Consecutive uppercase Latin letters (AI, KYC, UPI)
Hyphenated Pairs	Non-reduplicative hyphenated tokens (Start-Up, re-KYC)
Conjunctive Verbs	Absolute verb forms ending in -kar / -kē
Compound Verbs	Primary verb + vector/light verb (lēnā, dēnā, jānā)
Honorifics	Hindi T-V pronominal forms (āp, tum, tū)
MWEs	Fixed postpositional phrases (kē sāth, kē liyē, kē bād)

3.2 Regex Pipeline (Rule-Based)

The regex pipeline (pipeline.py) operates in four stages across four notional weeks:

- Week 1: Corpus exploration and UTF-8 validation — character/token statistics, sentence boundary detection using whitespace and danda markers.
- Week 2: Preprocessing and ISO 15919 transliteration — NFC normalization, custom Devanagari-to-Latin conversion with schwa deletion.
- Week 3: Pattern extraction — Reduplication (hyphen-split identical halves), Acronyms (regex `[A-Z]{2,}`), Hyphenated Pairs (non-reduplicative hyphenated tokens), Conjunctive Verbs (tokens matching `-(kar|ke)$`), Compound Verbs (any token followed within 3 tokens by a vector verb lemma), Honorifics (lexicon lookup of āp/tum/tū forms), MWEs (postposition sequence matching a fixed lexicon).
- Week 4: Integration of Social Media text — We added Social Media dataset and normalized it according to our need, manually transcribing when needed. We ran the pipeline built so far on this dataset and compared the results of the two modes against the two types of datasets.
- Week 5: Annotation of Gold Standard dataset — We created a gold standard with the annotations for all of the above metrics. We discussed ambiguities in order to ensure the most optimised gold standard.
- Week 6: Evaluation Metrics (Accuracy, Recall, Precision, F1 calculated) and Error Analysis — We evaluated the annotations received from the pipelines against the gold standard created and analysed the errors and found patterns from this.

The regex pipeline processes transliterated text exclusively. It is highly efficient, typically 50–200× faster than the neural pipeline, but is sensitive to transliteration errors and cannot leverage morphological context.

3.3 Neural Pipeline (Stanza)

The Stanza-based pipeline (neural_pipeline.py) uses the stanza Hindi model with four processors: tokenize, pos, lemma, and depparse. It operates directly on Devanagari text without transliteration, leveraging:

- POS tags (UPOS) for morphological filtering: VERB for conjunctive and compound verb detection, PRON for honorifics, ADP for MWEs.
- Lemmatisation for vector verb matching and honorific recognition across inflected forms (āpakō, āpanē, āpakī).
- Dependency structure (head pointer) for MWE extraction: an ADP postposition and its syntactic head are joined as a candidate MWE.

The neural pipeline is slower (~14 sentences/second vs 50–200× faster for regex) but morphologically aware, enabling recognition of inflected and partially occluded forms.

3.4 Manual Annotation and Evaluation

Both team members independently annotated approximately 725 sentences drawn randomly from both corpora, covering all seven patterns. Disagreements were resolved through discussion to produce a consensus gold standard. The gold standard is stored as JSON keyed by episode and pattern.

Precision, recall, and F1 were computed for each (system, corpus, pattern) combination. To avoid over-penalization of morphologically inflected variants, exact one-to-one string matching was relaxed to partial-overlap matching, allowing e.g. āpakō to match a gold annotation of āp [formal].

4. Results

4.1 Corpus Statistics

The formal corpus (broadcast) contains 9,763 tokens across 633 sentences; the informal corpus (comments) contains 4,230 tokens across approximately 112 sentences. All characters are valid Unicode with no encoding errors in either corpus.

4.2 Cross-Register Pattern Frequencies (Neural Pipeline)

The table below normalises raw pattern counts per 1,000 tokens to allow fair comparison between the differently-sized corpora.

Pattern	Formal (raw)	Formal /1k	Informal /1k	Ratio F:I
Reduplication	65	6.66	8.98	0.74×
Conjunctive Verbs	103	10.55	15.37	0.69×
Compound Verbs	118	12.09	19.15	0.63×
Honorifics	72	7.37	4.96	1.49×
Acronyms	33	3.38	1.42	2.38×
MWEs	124	12.70	4.73	2.69×
Hyphenated Pairs	67	6.86	0.71	9.67×
TOTAL	582	59.63	55.32	1.08×

4.3 Regex vs. Stanza Agreement — Formal Corpus

Pattern	Regex	Stanza	Stanza/Regex Ratio	Agreement
Reduplication	36	65	1.81×	Moderate
Conjunctive Verbs	91	103	1.13×	High
Compound Verbs	93	118	1.27×	High
Honorifics	25	72	2.88×	Low
Acronyms	33	33	1.00×	Perfect
MWEs	72	124	1.72×	Moderate
Hyphenated Pairs	21	67	3.19×	Low
TOTAL	371	582	1.57×	

4.4 Regex vs. Stanza Agreement — Informal Corpus

Pattern	Regex	Stanza	Stanza/Regex Ratio	Agreement
Reduplication	4	38	9.50×	Very Low
Conjunctive Verbs	34	65	1.91×	Moderate

Compound Verbs	53	81	1.53×	Moderate
Honorifics	15	21	1.40×	Moderate
Acronyms	6	6	1.00×	Perfect
MWEs	21	20	0.95×	Near-Perfect
Hyphenated Pairs	2	3	1.50×	Moderate
TOTAL	135	234	1.73×	

4.5 Overall Performance Metrics

System + Corpus	Precision	Recall	F1 Score	Accuracy
Neural — Broadcast	0.6146	0.4020	0.4861	0.8450
Regex — Broadcast	0.6321	0.2897	0.3973	0.8408
Neural — Comments	0.8806	0.8489	0.8645	0.9511
Regex — Comments	0.6579	0.5396	0.5929	0.8638

5. Per-Pattern Analysis

5.1 Reduplication

System	Corpus	F1	Precision	Recall
Neural	Broadcast	0.6392	0.544	0.775
Regex	Broadcast	0.2857	1.000	0.167
Neural	Comments	0.9474	0.900	1.000
Regex	Comments	0.9032	0.966	0.848

Regex reduplication on broadcast text is precision-perfect (1.000) but recall-devastated (0.167), finding only 3 of 18 annotated instances without a single false positive. On comments, both systems perform excellently (0.90+ F1). This inversion of the expected pattern — where formal text ought to be easier for regex — reveals that broadcast Hindi uses morphologically complex reduplicative forms (e.g. *kūṭ-kūṭ kar*, *kōṇē-kōṇē tak*) where the hyphen-split tokens do not have perfectly identical surface forms. Informal comments prefer simpler total reduplication (*mōdī-mōdī*, *har-har*) that the regex captures easily.

Crucially, the pragmatic function of reduplication shifts between registers: formal broadcast Hindi uses it as a grammatical distributive and adverbial device (*alag-alag* 'each separately', *dhīrē-dhīrē* 'gradually'), whereas informal social media Hindi repurposes the same form for rhetorical emphasis and political chanting (*mōdī-mōdī*, *āzādī-āzādī*).

5.2 Conjunctive Verbs

System	Corpus	F1	Precision	FP
Neural	Broadcast	0.4490	0.328	45
Regex	Broadcast	0.6316	0.581	13
Neural	Comments	0.4952	0.366	45
Regex	Comments	0.7097	0.611	14

Regex outperforms Stanza on conjunctive verbs in both registers. The decisive difference is precision: regex generates 13 false positives on broadcast versus Stanza's 45. The Stanza pipeline incorrectly tags many postpositions ending in *-ke* (e.g. *isa-kē* 'of this', *āpa-kē* 'your') as conjunctive verbs because their surface form matches the absolutive marker. Stanza achieves marginally higher recall in both

corpora but at the cost of more than three times the false positives. For downstream tasks where false positives are costly, regex is the recommended system for this pattern in all conditions.

5.3 Compound Verbs

System	Corpus	F1	TP	FN
Neural	Broadcast	0.4122	81	217
Regex	Broadcast	0.3680	62	203
Neural	Comments	0.9011	41	9
Regex	Comments	0.6353	27	23

This is the most counterintuitive result in the study. Both systems perform markedly worse on compound verbs in formal broadcast speech than in informal comments: the neural system drops from $F1=0.90$ to $F1=0.41$, and the regex from 0.64 to 0.37. The catastrophic false-negative counts (217 and 203 respectively) indicate that broadcast Hindi uses more complex, embedded compound verb constructions where the vector verb is separated from the primary verb stem by intervening material — a pattern neither system handles. Comments, by contrast, tend toward simple adjacent verb+auxiliary structures that both systems reliably identify.

5.4 Honorifics

System	Corpus	F1	Precision	FN
Neural	Broadcast	0.5507	0.864	56
Regex	Broadcast	0.2667	0.727	82
Neural	Comments	0.9697	1.000	1
Regex	Comments	0.6667	0.632	5

The neural pipeline dramatically outperforms regex on honorifics ($F1$: 0.55 vs 0.27 on broadcast; 0.97 vs 0.67 on comments). The primary cause is that broadcast Hindi uses *āp* almost exclusively, and in its many inflected forms — *āpakō* (dative), *āpanē* (ergative), *āpakī* (possessive), *āpakā* (possessive), *āpasē* (ablative) — which the regex pipeline's lexicon-based matcher does not capture. Stanza's lemmatiser normalises all inflected forms back to *āp*, enabling automatic recognition. The 56 and 82 false negatives for neural and regex respectively on broadcast suggest the gold standard annotates a broader honorific category than either system targets.

5.5 Acronyms

System	Corpus	F1	Precision	FN
Neural	Broadcast	0.8889	1.000	0.800
Regex	Broadcast	0.9057	1.000	0.828
Neural	Comments	0.7692	1.000	0.625
Regex	Comments	0.7692	1.000	0.625

Precision is perfect (1.000) across all four system-corpus conditions. Regex outperforms Stanza marginally on broadcast acronyms ($F1$: 0.906 vs 0.889) owing to slightly higher recall. Both systems fail

on a subset of gold-annotated acronyms that appear in mixed-case forms or embedded in longer tokens — a known limitation of the strict $[A-Z]\{2,\}$ boundary pattern. Acronyms are the strongest register marker in terms of precision and can be extracted with a simple rule.

5.6 Multi-Word Expressions (MWEs)

System	Corpus	F1	Precision	FP
Neural	Broadcast	0.2222	0.202	71
Regex	Broadcast	0.1417	0.184	40
Neural	Comments	1.0000	1.000	0
Regex	Comments	0.5294	0.450	11

The MWE results show the starkest performance contrast in the study. The neural pipeline achieves perfect MWE detection on comments (F1=1.00) but nearly fails on broadcast (F1=0.22, 71 false positives). The 71 false positives indicate that Stanza over-generates MWE candidates from the dense postpositional structure of formal Hindi: every ADP + head combination is a candidate, and formal discourse contains many more postpositional phrases than the lexicon-matched gold annotations capture. On informal comments, postpositional density is lower and the patterns are simpler, leading to perfect precision and recall. MWEs show a formal-to-informal frequency ratio of 2.69 $\times$, making them a reliable formal register marker when precision is acceptable.

5.7 Hyphenated Pairs

Hyphenated pairs exhibit the most extreme register sensitivity in the study, with a formal-to-informal ratio of 9.67 $\times$ (6.86 vs 0.71 per 1,000 tokens). Formal broadcast Hindi uses hyphenation systematically for English code-switching (Start-Up, re-KYC, under-16) and for geographic compounds (Jammu-Kashmir, Nāthu-Lā). Informal comments rarely hyphenate. The regex pipeline, which targets non-reduplicative hyphenated tokens with length ≥ 2 on each side, achieves higher precision than the neural pipeline on this pattern because the neural system captures Devanagari compound forms (nāḍī-kuśal) that the gold standard does not annotate as hyphenated pairs.

6. Cross-Register Analysis

6.1 Register-Stable Patterns

Compound verbs and conjunctive verbs are register-stable: both appear at rates per 1,000 tokens that are higher in informal comments than in formal broadcast (ratios of 0.63× and 0.69× respectively when expressed as formal/informal). This counter-intuitive result — informal speech using more verb-complex constructions — may reflect the absence of nominalized and passive constructions in informal discourse, which in formal registers absorb many semantic slots that informal registers express through serial verb strategies.

6.2 Register-Sensitive Patterns: Formal-Dominant

MWEs (2.69× formal/informal) and hyphenated pairs (9.67×) are strongly formal-dominant. MWEs reflect the elaborate postpositional syntax of broadcast Hindi, which employs constructions like *kē tahat* (under/pursuant to), *kē anusār* (according to), and *kī vajah sē* (because of) in bureaucratic and administrative discourse. Hyphenated pairs are largely absent from informal Hindi because English code-switching in social media comments tends to use bare English words without compounding.

Acronyms are moderately formal-dominant (2.38×): broadcast episodes discuss topics involving AI, KYC, GI tags, and NABARD, whereas comments refer only to political acronyms (RSS, RJD, IAS/IPS).

6.3 Register-Sensitive Patterns: Informal-Dominant

Reduplication appears slightly more frequently in informal comments per 1,000 tokens (8.98 vs 6.66, ratio 0.74×), but the more important difference is pragmatic: formal reduplication is grammatical (*alag-alag* as a distributive, *dhīrē-dhīrē* as a manner adverb), while informal reduplication is rhetorical (*mōdī-mōdī*, *har-har*, *āzādī-āzādī* as chant-forms). Honorifics show a formal-dominant ratio (1.49×) attributable to the broadcast genre's consistent use of *āp*-register second-person address to a national audience, while comments mix *āp* with *tum* (familiar) in interpersonal argument threads.

7. Error Analysis

7.1 Regex Error Patterns

- Overgeneralization of conjunctive verbs: postpositions ending in -ke (isa-kē, āpa-kē, tarī-kē) are incorrectly classified as absolutive verb forms. This accounts for the majority of false positives in both registers.
- Honorific miss: regex matches only the uninflected pronoun stems. Broadcast Hindi's near-exclusive use of inflected āp forms (āpakō, āpakī, āpanē, āpasē) results in catastrophic recall loss on the formal corpus.
- Compound verb false positives on transliterated text: the proximity-window heuristic (primary verb + vector within 3 tokens) fires on non-verb tokens that happen to precede vector verb forms, inflating the broadcast compound verb count.
- Reduplication on broadcast: the strict identical-halves heuristic misses morphologically complex reduplications (kūṭ-kūṭ, kōnē-kōnē) where the hyphenated tokens have different case suffixes.

7.2 Neural (Stanza) Error Patterns

- Overgeneralization of reduplication: Stanza's lemma-matching heuristic catches non-reduplicative pronoun pairs (vō-usa, vē-una) that have identical lemmas but different surface referents. This inflates recall at the cost of precision on broadcast.
- MWE false positives on broadcast: the dependency-based MWE extractor produces 71 false positives by flagging every ADP + head pair, including compositional postpositional phrases that are not fixed expressions.
- Conjunctive verb false positives: Stanza generates 45 false positives in both registers, compared to regex's 13–14, by incorrectly tagging postpositions and certain nominal forms as VERB.
- Code-switched English token lemmatisation: Stanza's Hindi lemmatiser fails on English tokens (Start-Up → start-up, re-KYC → re-kyc), degrading hyphenated pair detection.

7.3 Shared Limitations

- Compositional MWEs: both systems fail to distinguish lexicalized fixed expressions (kē sāth as 'together with') from compositionally transparent postpositional phrases (kē sāth as 'with [that person]'). Distinguishing these requires semantic context.
- Embedded compound verbs in broadcast: both systems miss compound verb constructions in which an adverb, negation, or object intervenes between the primary verb and the vector verb, a construction more frequent in formal than informal Hindi.
- Gold standard alignment: the relaxed matching approach mitigates but does not eliminate misalignment between gold annotations (often at the sentence level) and system outputs (at the token or phrase level).

8. Challenges and Mitigations

Challenge	Mitigation
No readily available informal Hindi corpus of adequate size (HASOC and Dakshina datasets lacked sufficient informal content).	Scraped YouTube comments on Mann Ki Baat videos; ran a normalization pipeline to remove emojis and non-Hindi/English content.
Ambiguous pattern boundaries: conjunctive verbs vs postpositions in -ke, compound verbs vs simple verb sequences, compositional vs lexicalized MWEs.	Discussed and reached consensus definitions for each ambiguity before annotation; applied those definitions consistently in the gold standard.
Schwa deletion: medial schwas remain undeleted (sarakār vs sarkār), complicating rule-writing.	Adopted a uniform convention: schwa deletion is word-final only throughout the pipeline.
Metrics over-penalization: exact string matching was too rigid, assigning 0 credit to morphologically inflected variants of a correctly identified pattern.	Relaxed matching: a predicted token counts as TP if it overlaps with or is a morphological variant of any gold annotation in the same sentence.
Stanza's lemmatiser unreliable on code-switched English tokens.	Noted as a known limitation; English tokens were excluded from lemma-dependent extraction steps.

9. Conclusion

9.1 Summary of Findings

This study provides an empirical evaluation of regex and neural approaches to Hindi linguistic pattern extraction across two registers. The principal findings are:

- Neural (Stanza) outperforms regex overall, but the gap is smaller than expected: regex achieves F1=0.40 on broadcast and F1=0.59 on comments, compared to Stanza's F1=0.49 and F1=0.86 respectively.
- Regex counts remain reliably proportional to neural outputs across patterns and registers, validating regex as an efficient proxy for cross-register linguistic analysis.
- MWEs and hyphenated pairs are strong register markers (formal-to-informal ratios of 2.69× and 9.67×) detectable with high precision by simple rules.
- Reduplication is a misleading surface-stable pattern: while raw frequency ratios are modest, the pragmatic function shifts dramatically between registers, grammatical device in broadcast, rhetorical chant in comments.
- Regex outperforms Stanza on conjunctive verbs in both registers (F1: 0.63 vs 0.45 on broadcast, 0.71 vs 0.50 on comments), confirming that for morphologically regular, surface-predictable patterns, rule-based methods can remain competitive on noisy real-world text.
- Compound verbs in formal broadcast Hindi exceed both systems' capacity, with catastrophic false-negative counts (200+) indicating that embedded, non-adjacent constructions require more sophisticated dependency-aware extraction.

9.2 Implications

For low-resource or real-time NLP applications on Hindi text, the choice of system should be pattern-specific. Acronyms and hyphenated pairs need only a simple regex rule. Conjunctive verbs favor regex.

Compound verbs and honorifics require neural support, particularly on formal text. MWEs are best extracted with a hybrid approach: neural on informal text ($F1=1.00$), lexicon-based regex on formal text where neural over-fires.

The register sensitivity findings suggest that a pattern-level frequency profile, computed cheaply with regex, can classify Hindi text by register without a trained classifier, at least for MWEs and hyphenated pairs whose distributional divergence is large enough to be discriminative.

9.3 Future Work

- Expand the gold standard to 1,500+ sentences to reduce annotation variance on low-frequency patterns.
- Implement a dependency-aware compound verb detector that tolerates intervening material between primary verb and vector verb.
- Extend the MWE lexicon to cover a broader class of postpositional constructions and test with a precision-recall threshold to filter over-generation.
- Apply the pattern-frequency profile as a feature vector for a register classifier and evaluate its separability against supervised baselines.
- Extend the corpus to additional Mann Ki Baat episodes and additional social media platforms to improve generalization.

10. Contributions

Member	Contributions
Ananya Pai	Unicode conversion and ISO 15919 transliteration module; Stanza-based neural pipeline (<code>neural_pipeline.py</code>); data annotation for social media corpus; computation of precision/recall/F1 metrics against gold standard for both pipelines.
Vidushi Agarwal	Dataset selection (broadcast and informal corpora); regex pipeline (<code>pipeline.py</code> , Weeks 1–4); informal dataset scraping and normalization; discussion of annotation conventions for social media text.
Joint	Independent annotation of ~725 sentences; consensus discussion to resolve disagreements; co-authorship of this report.

11. References

- Qi, P., Zhang, Y., Zhang, Y., Bolton, J., & Manning, C. D. (2020). Stanza: A Python natural language processing toolkit for many human languages. In Proceedings of ACL 2020 System Demonstrations.
- Kellogg, S. H. (1893). A Grammar of the Hindi Language. London: Kegan Paul. (Reference for conjunctive verb and compound verb definitions.)
- Gupta, D., Ekbal, A., & Bhattacharyya, P. (2020). A deep neural network framework for English-Hindi code-mixing. ACM Transactions on Asian and Low-Resource Language Information Processing.
- Singh, A. K., & Gorla, J. (2007). Identification of languages and encodings in a multilingual document. In Proceedings of the 3rd ACL SIGWAC Workshop on Web As Corpus.
- Kunchukuttan, A., Mehta, P., & Bhattacharyya, P. (2018). The IIT Bombay English–Hindi Parallel Corpus. LREC 2018.

Appendix A: Top Pattern Examples

A.1 Reduplication — Formal Broadcast (Regex, transliterated)

- alag-alag (8×) — 'separately/each'
- bahut-bahut (4×) — 'very much'
- apanē-apanē (3×) — 'each one's own'
- kōnē-kōnē (3×) — 'in every corner'
- samay-samay (3×) — 'from time to time'

A.2 Reduplication — Informal Comments (Stanza, Devanagari)

- बड़े-बड़े (3×) — 'very big / big people'
- मोदी-मोदी (2×) — chant form
- आजादी-आजादी (2×) — political slogan
- साथ-साथ (2×) — 'together'
- बहुत-बहुत (2×) — 'very much'

A.3 Top MWEs — Formal (Regex)

- kē sāth [with/along with] — 35×
- kē liyē [for/in order to] — 7×
- kē bīc [between/among] — 7×
- kē bād [after] — 4×
- sē lēkar [from...up to] — 4×

A.4 Top Acronyms — Formal

- AI — 15× (highest in February 2026 episode on Global AI Summit)
- KYC — 6×
- GI — 2× (Geographical Indication tags)
- UPI, SMS, OTP, NABARD — 1× each